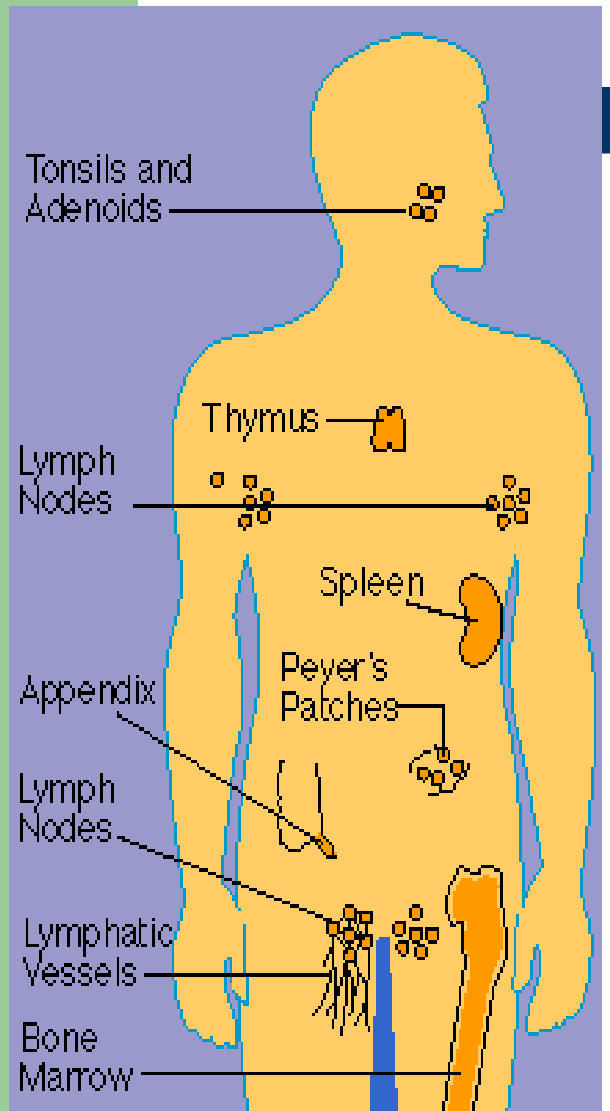
# Organs and cells of immune system

*Prof. Dr. Ruqaya M. Al Barzinji*
*Hawler Medical University*
*College of Medicine*
*Department: Microbiology*
*SGL 1:*

# Organs of the Immune System:-




- **1-Primary lymphoid organs:**

They are the major sites of lymphopoiesis.

- There are **two** major primary lymphoid organs, one in which the **T cell** develop and the other in which the **B cells** develop.

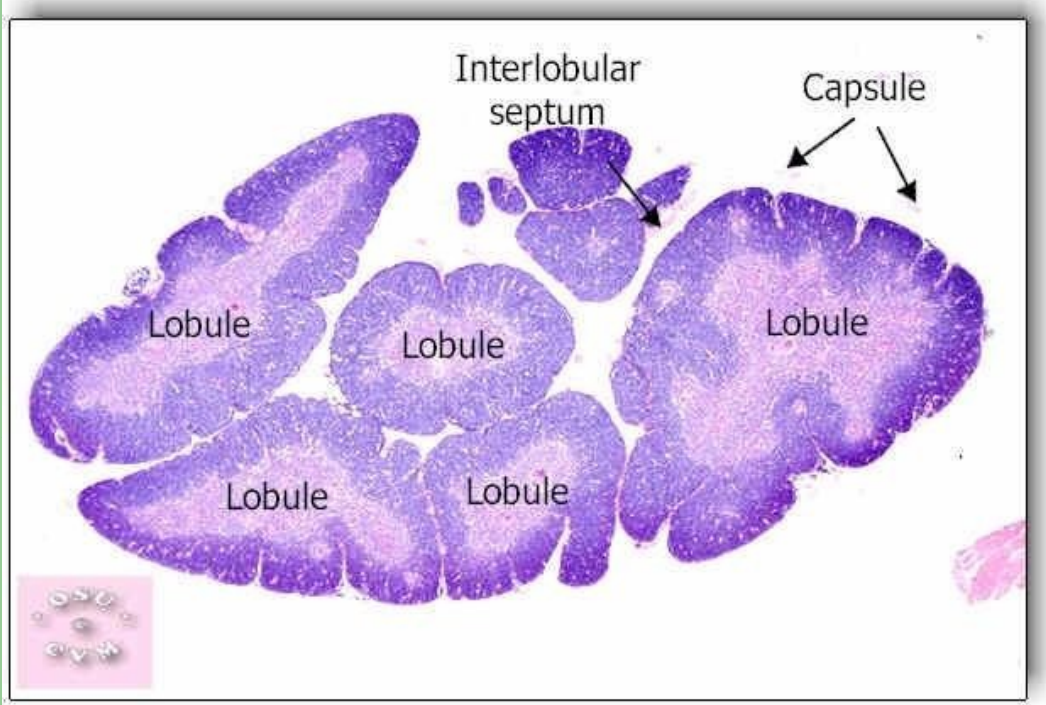
# 1- Thymus gland:-

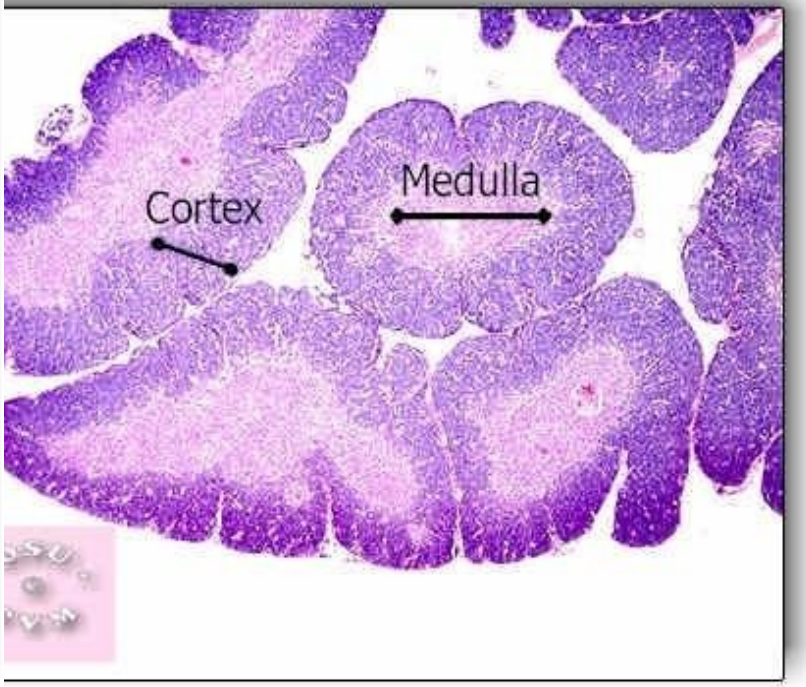
- It is a lymphoepithelial, bilobed structure.
- During fetal development, the size of the thymus increases until puberty and then slowly atrophies.
- **Types of cells in the thymus:-**

1 Thymocytes, morphologicaly similar to blood lymphocytes

2 Phagocytic reticulum

3 Reticular epithelial cells




- The thymus is organized into **outer cortical area** and **central medullary areas** that infiltrated with lymphoid cells (thymocytes).
- The cortex is densely populated with **epithelial cells**, **lymphocytes** of various sizes, and scattered **macrophages** involved in clearing apoptic thymocytes.

# Maturation of T lymphocytes:-

- The maturation of T lymphocytes occurs mainly during **fetal development** and for a short time after birth,
- They **mature in the cortex** and migrate to the medulla where they encounter macrophage and dendritic cells(rich in MHC class II) and undergo **thymic selection** which **eliminates self-reactive** and results in the development of mature T-cells.

# Thymic selection

- 1-Early in the development thymocytes they are lacking CD4 or CD8 molecules that is why called **double negative thymocytes**.
- **Rearrangement of T cell receptor**(TCR) genes causes co-expression of **CD3 and TCR αβ-heterodimer** at low level on the double negative cell surface.
- At this level, the **CD4 and CD8** both are **expressed** and the thymocytes are called **double positive**.

# Thymic selection

- In the subsequent development the level of TCR expression increases and cells **loose expression** of either **CD4 or CD8** and become **single positive functional T cells**
- which then leave the thymus through **post-capillary venules** to enter the peripheral blood circulation through which they are transported to the secondary lymphoid organs in which T cells encounter and respond to foreign antigens.
- This recognition is achieved by a specific TCR which is acquired during differentiation in the

# 2-Bone marrow:-

- A primary lymphoid organs (bursa of Fabricious) was first discovered in **birds**, near the cloaca which B cells undergo maturation in it.
- In **mammalian** bone marrow is equivalent organ.
- The immunoglobulin **genes** are rearranged as the cell differentiates and the mature B-cell leaves the bone marrow with an antigen receptor(membrane immunoglobulin) ready to respond to antigen if necessary.

- The mature B cells are transported by the circulating blood to the secondary lymphoid organs, where they encounter and responded to foreign antigens.

## Secondary lymphoid organs:-

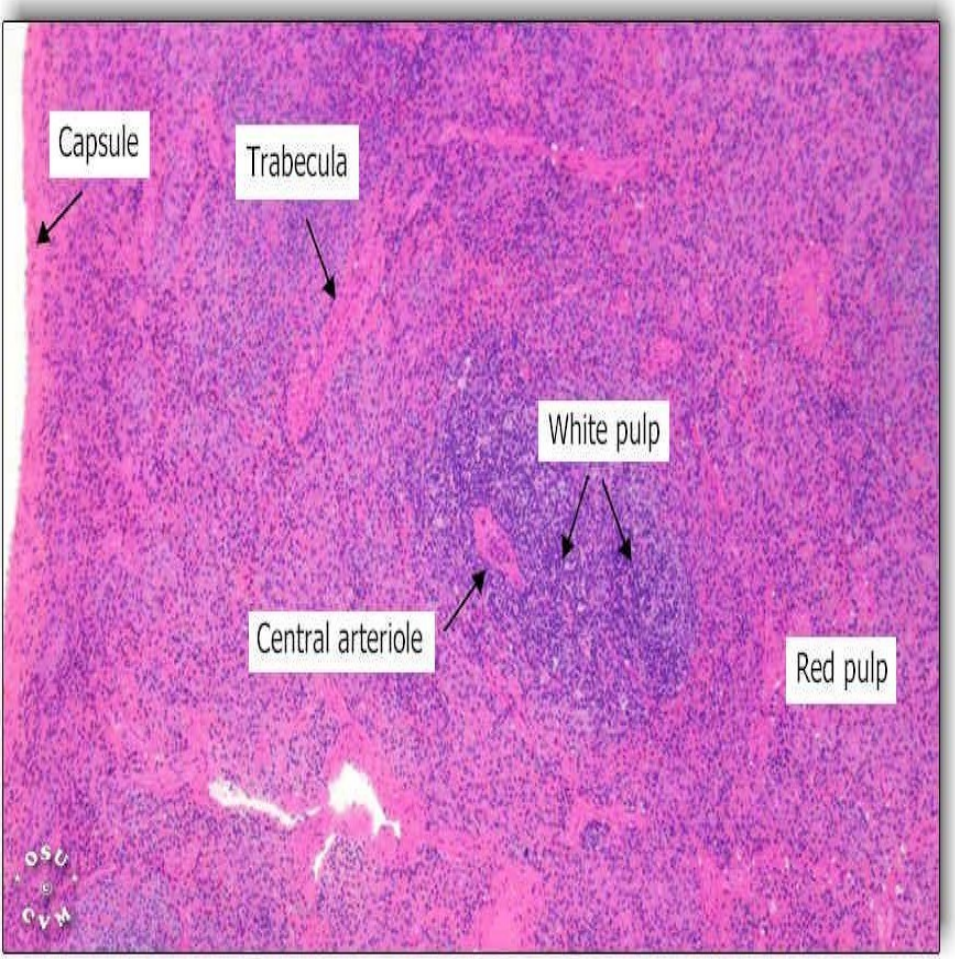
- Secondary lymphoid organs includes:
- **lymph nodes**, **spleen**, **mucosal associated** lymphoid tissue(MALT) as gut-associated lymphoid tissue(GALT),bronchus associated lymphoid tissue(BALT),tonsils and Payers patches of the gut.

# The major functions of secondary lymphoid organs:-

1 They are highly efficient in trapping and concentrating foreign substances.

2 They are the main sites of **production of antibodies** and the **induction** of antigen-specific T lymphocytes.

# 1-The spleen:-




- The spleen is the largest of the secondary lymphoid organs.
- It is enclosed by a capsule and divided by trabeculae into communicable compartments.

# 1-The spleen:-

- 1-It is composed of white pulp, rich in lymphoid cells and located around small arteriols. Red pulp which contains many sinuses as well as large quantities of erythrocytes, macrophage and some lymphocytes.
- 2-Approximately **50% of spleen cells are B** lymphocytes and **30-40% are T-lymphocytes**.

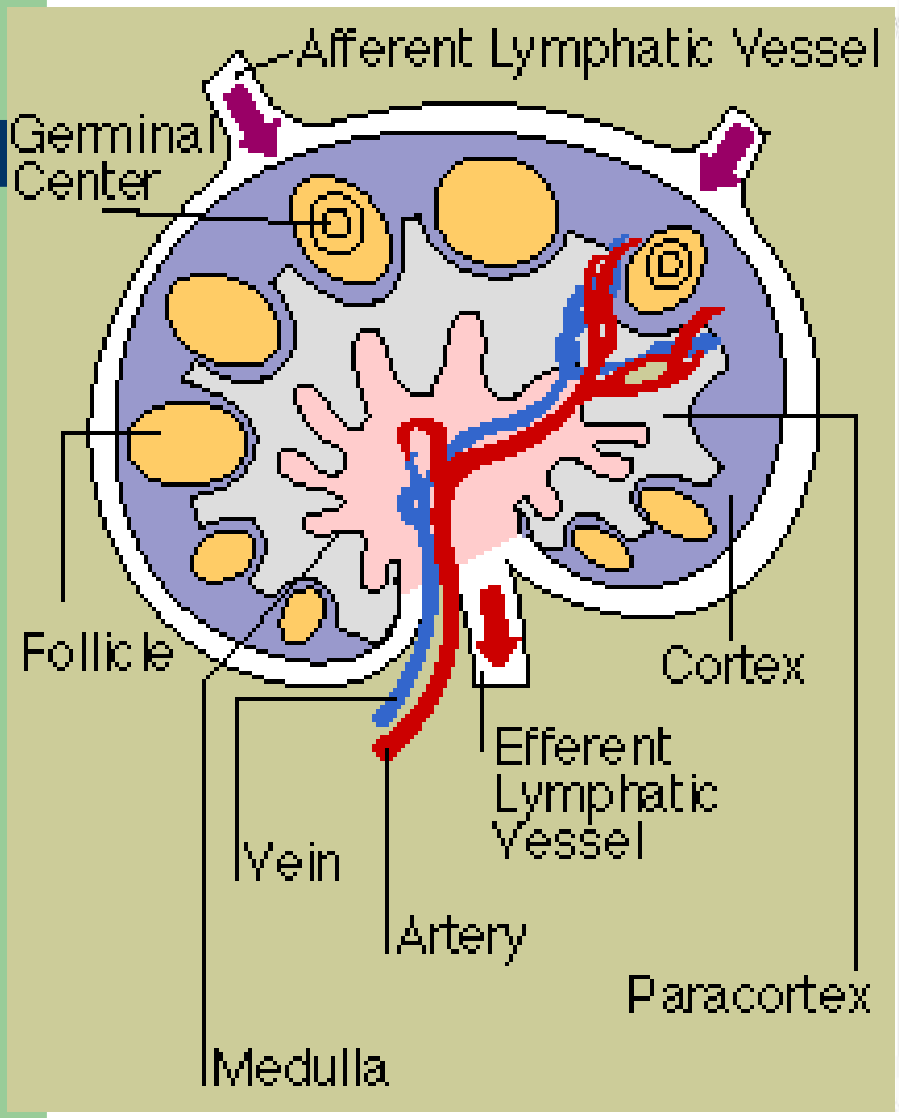
- 3- It is highly efficient in trapping and concentrating foreign substances carried in the blood.
- 4-It is the major organ in the body in which antibodies are synthesized and from which they are released into the circulation.

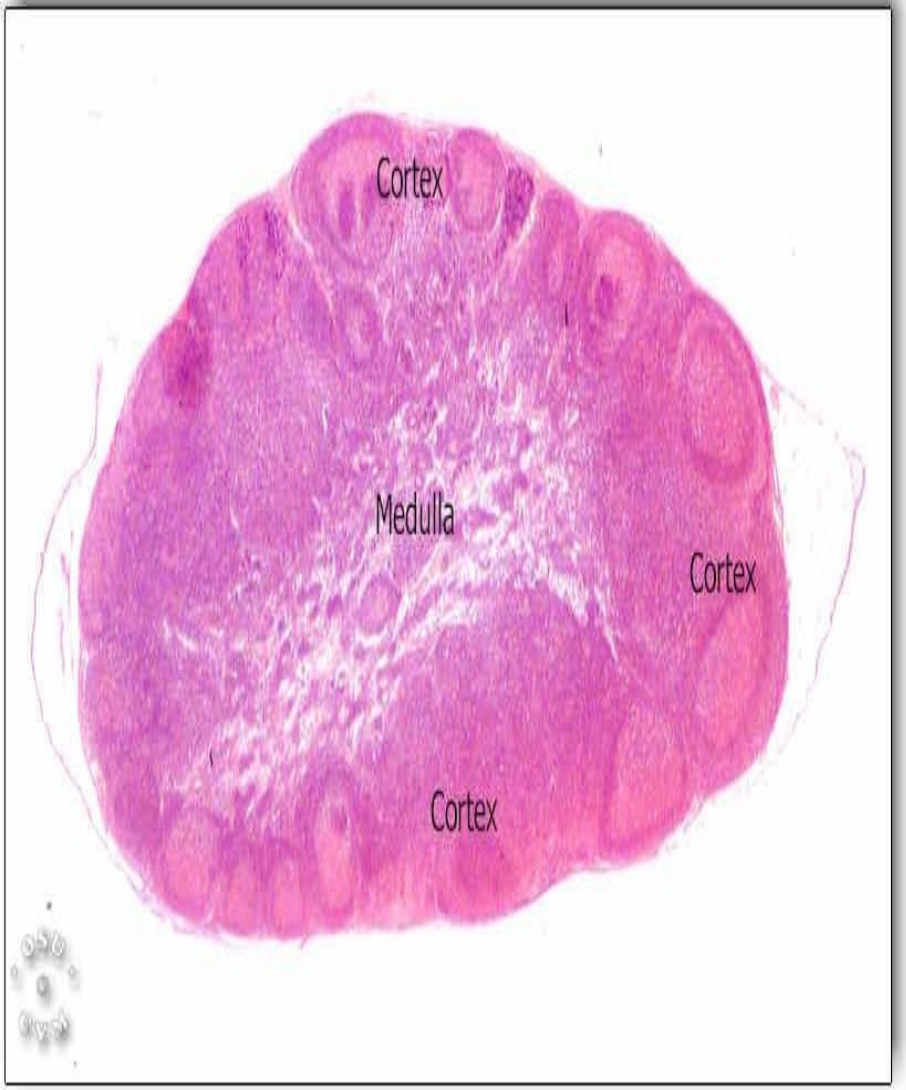
# 2-Lymph nodes:-

- Lymph nodes are small, bean-shaped structures that are laced throughout the body along the lymphatic routes.
- Lymph nodes contain specialized compartments where immune cells congregate, and where they can encounter antigens.

# 2-Lymph nodes:-

- The lymph nodes are composed of **medulla and cortex** which surrounded by a capsule of connective tissue.
- The cortical region contains primary lymphoid follicles, after antigenic stimulation enlarge to form secondary follicles with germinal center rich with B cells.
- The paracortical area contains T cells and dendritic cells, antigen presenting occur that result in T cell activation. The medullary area


Afferent Lymphatic Vessel
Germinal Center
Follicle
Cortex
Efferent Lymphatic Vessel
Vein
Artery
Paracortex
Medulla


Cortex
Medulla
Cortex
Cortex

- Lymphocytes enter the lymph node via afferent lymphatics or from the blood through specialized cells called high wall endothelium(HEV) which found in the post – capillary venule that present in cortex and paracortex.

# The major function of lymph nodes:-

- 1-Lymph nodes are highly efficient in trapping antigen that enters through the afferent lymphatic vessels.
- 2- In the node the antigen interact with macrophage, T cells and B cells and that interaction brings about an immune response represented by generation of antibodies and antigen-specific T cells.

# Cells of immune system can be classified into:-

- **1-Myloied linage**
- **2- Lymphoid linage:-**
- **The myeloid linage include:-**
- **A-Mononuclear phagocytes**: They are found in the blood and in tissues. In the blood they are known as **monocytes** while in the tissues they differentiate into different histologic forms for example

- 1-In connective tissue they are known as histocytes
- 2-In the kidney as measangial cells
- 3- In bone as osteoclasts
- 4-In brain as microglia
- 5- In the spleen, lymph node and thymus as the sinus lining macrophage
- 6- In the liver as Kupffer cells
- 7- In the lungs as alveolar macrophage
- 8-Peritonial macrophage, free floating in peritoneal fluid

# B-Multinuclear phagocyte:

- Polymorphonuclear cells(PMN) are include Neutrophil, eosinophil and basophile they kill the target by increase phagocidal activity mechanisms.

# The lymphoid linage include:-

- A-Small a granular lymphocyte
- 1-B-lymphocyte (B-cell)
- 2- T-lymphocyte (T-cell), sub classified into
- A: T-helper (Th-cell)
- B: T-cytotoxic (Tc-cell)
- C: T-suppressor (Ts-cell)
- **B-Large granular lymphocyte**
- **Natural killer cells (NK cells):-**NK cells are large granular lymphocytes, they play a role in the early stages of viral infection and tumorogenesis.

Thanks a lot